Practical 8: MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements Problem Definition

Design the database for StudentHub with students, events and registrations. Create PHP database connection using MYSQL statements.

Create the database and tables

Open http://localhost/phpmyadmin in your browser. Click the SQL tab and execute this SQL code.

```
CREATE DATABASE student_hub;

USE student_hub;

CREATE TABLE students (
    student_id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    course VARCHAR(50) NOT NULL
);

CREATE TABLE events (
    event_id INT AUTO_INCREMENT PRIMARY KEY,
    event_name VARCHAR(100) NOT NULL,
```

```
    event_date DATE NOT NULL,
    venue VARCHAR(100) NOT NULL
);

CREATE TABLE registrations (
    registration_id INT AUTO_INCREMENT PRIMARY KEY,
    student_id INT NOT NULL,
    event_id INT NOT NULL,
    registration_date DATE NOT NULL,
    FOREIGN KEY (student_id) REFERENCES
students(student_id),
    FOREIGN KEY (event_id) REFERENCES events(event_id),
    UNIQUE (student_id, event_id)
);
```

Step 2: Create the PHP database connection

In your XAMPP project folder, create a file named db.php.

For example, save it at D:\xampp\htdocs\student-hub\db.php if your XAMPP installation is on the D: drive.

```
<?php
$host = "localhost";
```

```
$user = "root";
$password = "";
$database = "student_hub";

$conn = mysqli_connect($host, $user, $password,
$database);

if (!$conn) {
   die("Connection failed: " . mysqli_connect_error());
}

echo "Database connected successfully";
?>
```

HTML:-

```
<!DOCTYPE html>
<html>
<head>
   <title>StudentHub</title>
</head>
<body>
   <h1>StudentHub</h1>
```

```
<h2>Student Registration Form</h2>

<form action="register.php" method="POST">
    <label>Student Name:</label>
    <input type="text" name="name" required>
    <br><br>

    <label>Email:</label>
    <input type="email" name="email" required>
    <br><br>

    <label>Course:</label>
    <input type="text" name="course" required>
    <br><br>

    <input type="submit" value="Register">
</form>

<hr>

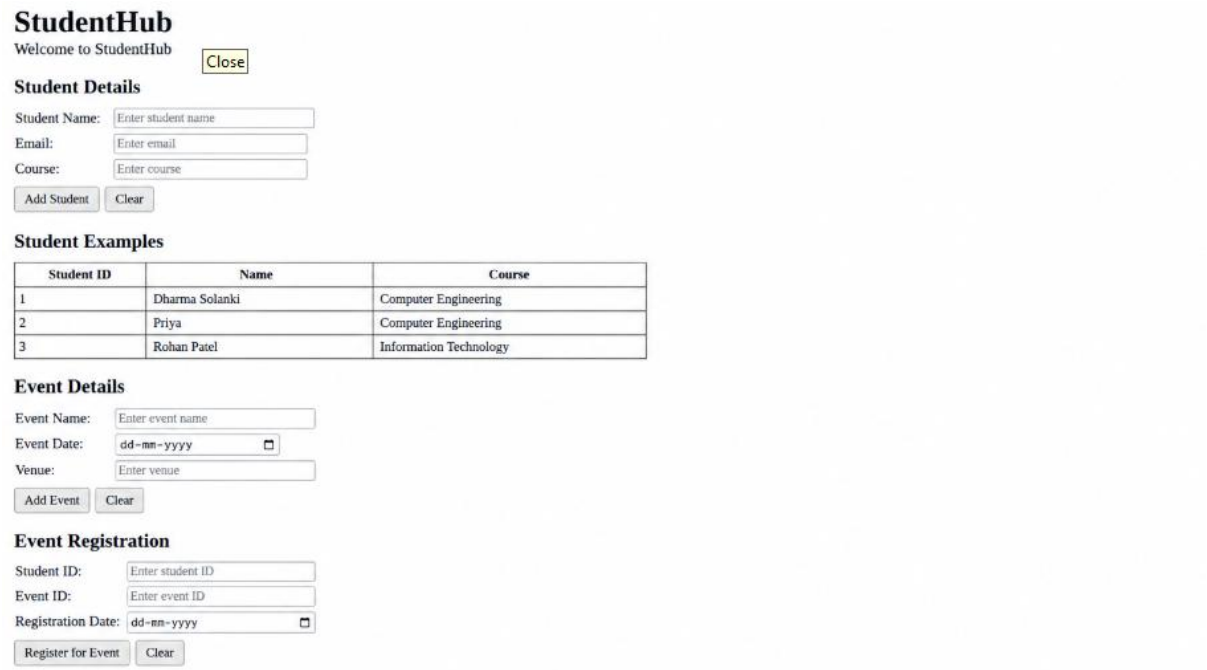
<h2>Event Registration</h2>
```

```
    <form action="event_register.php" method="POST">
        <label>Student ID:</label>
        <input type="number" name="student_id" required>
        <br><br>

        <label>Event ID:</label>
        <input type="number" name="event_id" required>
        <br><br>

        <input type="submit" value="Register for Event">
    </form>
</body>
</html>
```

Output:-



student_id	name	email	course
1	Dharma Solanki	dharma@gmail.com	Computer Engineering
2	Priya	priya@gmail.com	Computer Engineering
3	Rohan Patel	rohan@gmail.com	Information Technology

Key Questions / Analysis / Interpretation

1. Is the schema normalized properly?

Yes, the StudentHub database is normalized by separating data into three tables: students, events, and registrations. This reduces data duplication and maintains data consistency.

2. Are primary keys and foreign keys correctly defined?

Yes, each table has a primary key (student_id, event_id, and registration_id). The registrations table contains foreign keys student_id and event_id to establish relationships between students and events.

3. Is the database connection handled securely?

The database connection is handled using PHP MySQLi. Connection errors are checked using mysqli_connect_error(). For better security, prepared statements should be used for database queries, and database credentials should be stored securely.

Tools/Technology to be Used MySQl

phpMyAdmin, PHP, XAMPP,VS code